

# MAC SPOOFING ATTEMPT IN WINDOWS AND KALI LINUX

## **Disclaimer:**

- This project is done for educational purpose only and not to be used for any malicious activity.

## **Self-Lab Descriptions:**

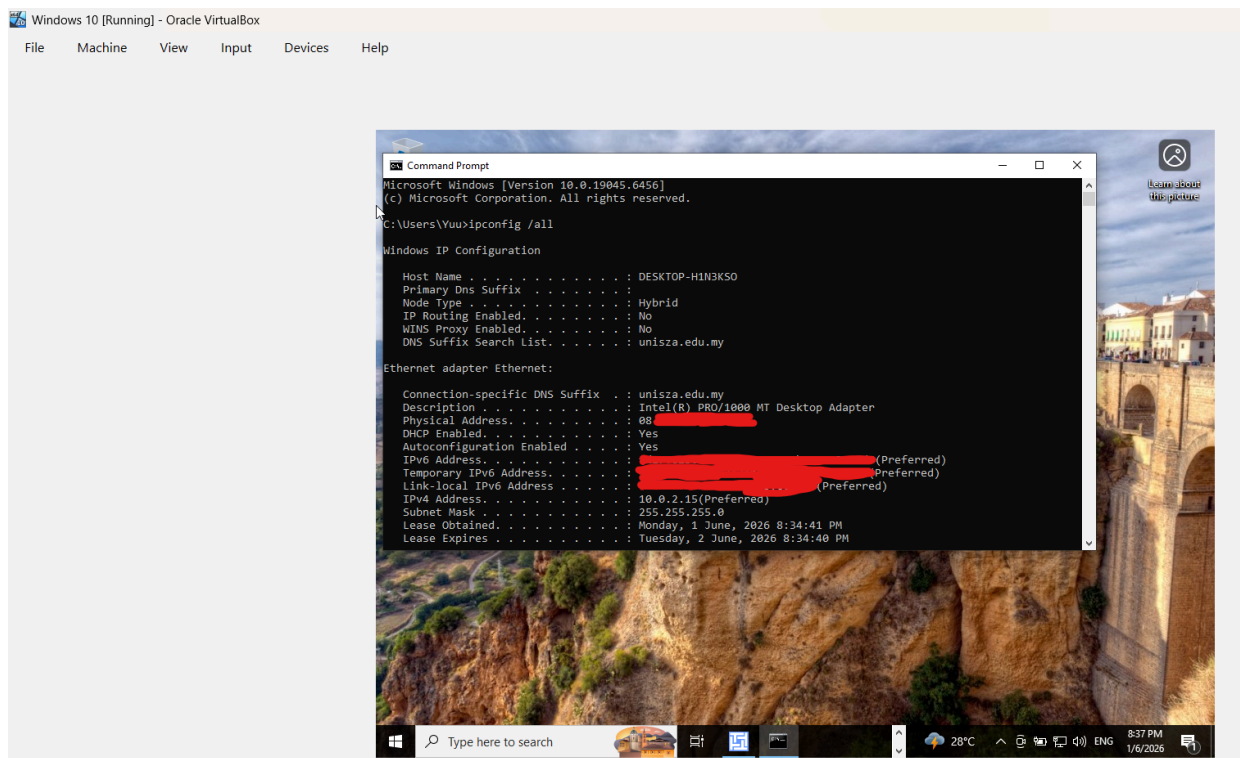
- To demonstrate IP spoofing technique through changing a machine's MAC address in Windows and Kali Linux.
- Tools used: Windows 10 iso, Kali Linux, Virtual Box to run the 2 virtual machines and Technitium MAC Address Changer to change the MAC Address in Windows.

## **WINDOWS Virtual Machine**

Open Windows terminal@command prompt.

Before changing the MAC address, check the original first through typing *ipconfig /all* in the terminal. It is to note that in Windows, MAC address is labelled as Physical Address.

For security reasons, the full mac address and IPv6 infos are to be disclosed. Here we can see 08-x-x-x-x-x.



**Figure 1: Original MAC Address**

Technitium is then used to change the MAC Address.

Open Technitium > Click *IP Address* > Choose *Random MAC Address* > Click *Apply*.

Close the Terminal and re-open it again. Then type *ipconfig /all*.

Now we can see the Physical (MAC) Address has changed and is the same one as shown in Technitium Ethernet's MAC Address table.

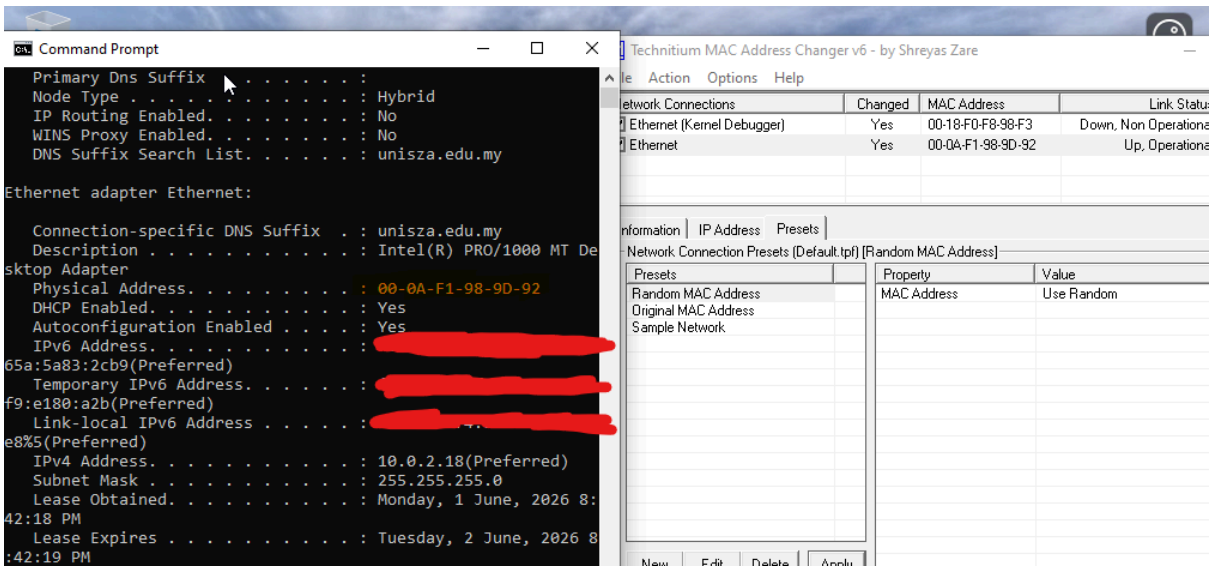


Figure 2: Changed MAC Address

End for Windows Lab

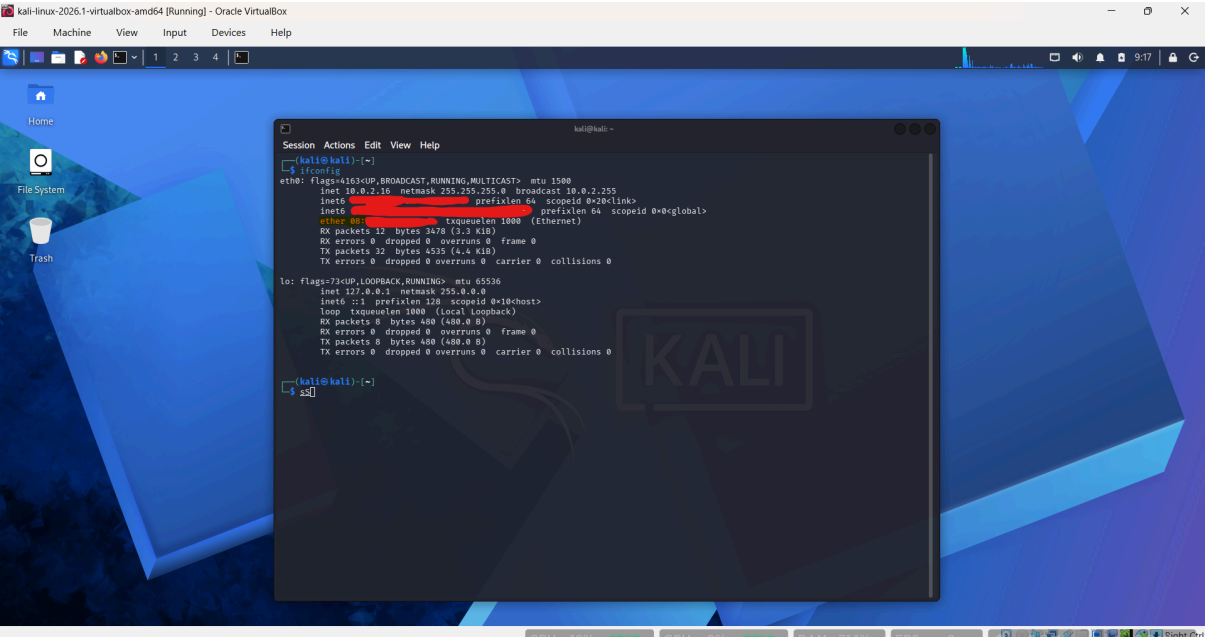
## KALI LINUX Virtual Machine

Unlike Windows that require an application to change MAC Address, in Kali Linux, we can simply use the Terminal.

The procedure is as below:

1. Disable the wanted interface.
2. Change the MAC Address.
3. Enable the interface.

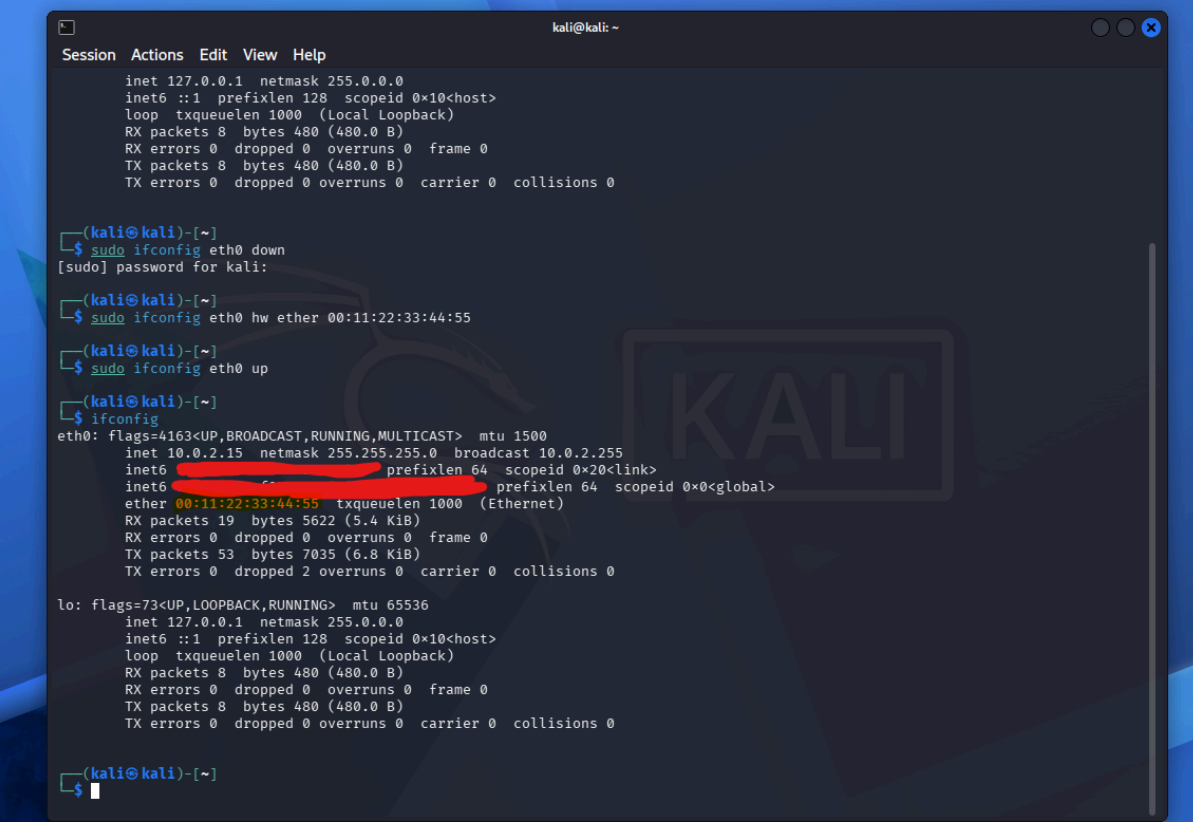
First we checked the currency MAC Address via typing *ifconfig*. Pick the one next to *ether*. Here, again, we see *08:x:x:x:x:x*.



```
kali@kali: ~  
$ ifconfig  
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 10.0.2.16 netmask 255.255.255.0 broadcast 10.0.2.255  
    inet6 ::1 prefixlen 64 scopeid 0x20<link>  
    ether 08:00:27:16:00:00 txqueuelen 1000 (Ethernet)  
    RX packets 12 bytes 3478 (3.3 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 32 bytes 4535 (4.4 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 0 bytes 400 (400.0 B)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 8 bytes 488 (488.0 B)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
kali@kali: ~  
$
```

Figure 3: Default MAC Address

We will choose the *eth0* interface. Proceed to disable the interface by typing *sudo ifconfig eth0 down*.



```
kali@kali: ~  
Session Actions Edit View Help  
inet 127.0.0.1 netmask 255.0.0.0  
inet6 ::1 prefixlen 128 scopeid 0<10<host>  
loop txqueuelen 1000 (Local Loopback)  
RX packets 8 bytes 480 (480.0 B)  
RX errors 0 dropped 0 overruns 0 frame 0  
TX packets 8 bytes 480 (480.0 B)  
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
(kali@kali)-[~]  
$ sudo ifconfig eth0 down  
[sudo] password for kali:  
(kali@kali)-[~]  
$ sudo ifconfig eth0 hw ether 00:11:22:33:44:55  
(kali@kali)-[~]  
$ sudo ifconfig eth0 up  
(kali@kali)-[~]  
$ ifconfig  
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255  
inet6 [REDACTED] prefixlen 64 scopeid 0<20<link>  
inet6 [REDACTED] prefixlen 64 scopeid 0<0<global>  
ether 00:11:22:33:44:55 txqueuelen 1000 (Ethernet)  
RX packets 19 bytes 5622 (5.4 KiB)  
RX errors 0 dropped 0 overruns 0 frame 0  
TX packets 53 bytes 7035 (6.8 KiB)  
TX errors 0 dropped 2 overruns 0 carrier 0 collisions 0  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
inet 127.0.0.1 netmask 255.0.0.0  
inet6 ::1 prefixlen 128 scopeid 0<10<host>  
loop txqueuelen 1000 (Local Loopback)  
RX packets 8 bytes 480 (480.0 B)  
RX errors 0 dropped 0 overruns 0 frame 0  
TX packets 8 bytes 480 (480.0 B)  
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
(kali@kali)-[~]  
$
```

**Figure 4: Changed MAC Address**

Then change the MAC Address by typing *sudo ifconfig eth0 hw ether 00:11:22:33:44:55*.

**00:11:22:33:44:55** is our fake MAC Address.

Then enable the interface by typing *sudo ifconfig eth0 up*.

Now we check the current MAC Address in the Terminal by typing *ifconfig*.

Attempt successful.

**End of Kali Linux Lab**